nanoker

CERAMIC TECHNICAL DATA SHEET

KBNC

Boron Nitride Composite

KEY PROPERTIES

- High resistance to thermal shock
- Good machinability
- High electrical insulation
- Good corrosion and wear resistan-ce
- Non-wetted by non-ferrous molten metals

APPLICATIONS

- Non-wetted by non-ferrous molten metals
- Insulators for high temperature furnaces
- Wall channels for electrical propul-sion systems

MAIN PROPERTIES

Property	Value	Unit
Density	3.0	g/cm^3
Flexural strength	194	MPa
Hardness (knoop)	120	kg/mm^2
Thermal Conductivity (XY)	35	W/m·K
Thermal Conductivity (Z)	22	W/m·K
CTE (XY)	5.1	$x10^{-6}\ K^{-1}$

** All properties measured at 20ªC unless othervise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1800	ºC
Recomended temp	<1700	ºC
Continuous temp	1600	ºC

ELECTRICAL BEHAVIOR

Insulator	
Dielectric performance	High

MATERIAL INFORMATION

Material type	Boron Nitride composite
Structure	Layered ceramic
Machinability	Good
Key feature	Thermal shock resistance
Color	Light grey

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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